

Algebra II with Trigonometry

Chapter 6.2

Olympiad Problems (GPT-5.4)

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[4506907_alg-2trig-h-chp-6-2-note-docx.pdf](#)

Problems

1. Evaluate exactly:

$$\sin \frac{13\pi}{12} + \cos \frac{13\pi}{12}.$$

2. A real number t satisfies

$$\sin t + \cos t = \frac{1}{3}.$$

Knowing t is in Quadrant II, find the exact value of $\tan t$.

3. Without a calculator, determine the exact value of

$$\sec \frac{7\pi}{12} - \csc \frac{7\pi}{12}.$$

4. If

$$\cos t = -\frac{5}{13} \quad \text{and} \quad \sin 2t < 0,$$

find the exact value of

$$\tan t + \cot t.$$

5. Let

$$y(t) = 3 \cos(\pi t) + 4 \sin(\pi t).$$

Find the largest possible value of $y(t)$, and one value of $t \in [0, 2)$ at which it occurs.

6. Find the exact value of

$$\left(\sin \frac{\pi}{12}\right) \left(\cos \frac{5\pi}{12}\right) + \left(\cos \frac{\pi}{12}\right) \left(\sin \frac{5\pi}{12}\right) - \tan \frac{3\pi}{4}.$$